

Labb 1: Driven damped pendulum

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Q1: Period of Undamped Pendulum

Method: We calculated the period T of oscillation for different initial angular displacements θ_0 by simulating the differential equation $\ddot{\theta} + \omega_0^2 \sin(\theta) = 0$ using the 4th order Runge-Kutta (RK4) method. The simulation detected turning points to find the half-period. We compared this to a small angle approximation, a power series expansion, and direct numerical integration.

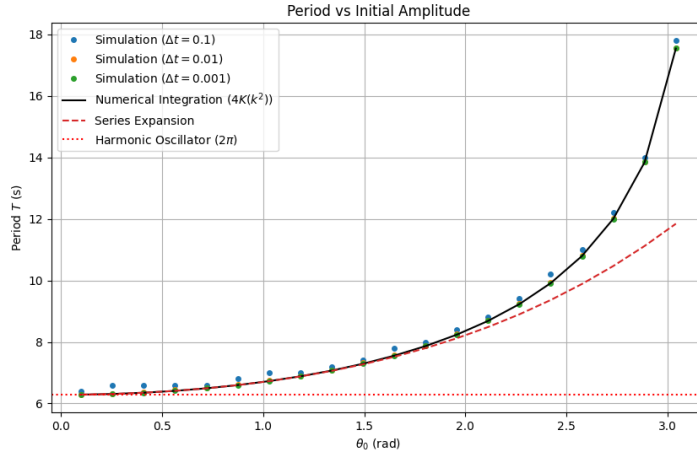


Figure 1: Period T as a function of the initial angular displacement θ_0 . Parameters: $\theta_0 \in [0.1, \pi - 0.1]$ (20 linearly spaced values), $\omega_0^2 = 1$. RK4 time steps: $\Delta t \in \{0.1, 0.01, 0.001\}$.

Theory: For small initial angles, the pendulum behaves as a simple harmonic oscillator with a constant period $T = 2\pi$ (when $\omega_0^2 = 1$). For larger angles, the restoring force $\sin(\theta)$ is weaker than the linear approximation θ , which implies the period should increase with amplitude. The power series expansion $T \approx 2\pi(1 + \theta_0^2/16 + \dots)$ accounts for this but will eventually diverge from the true solution for very large angles.

Results: As shown in Figure 1, all methods agree and yield $T \approx 2\pi$ for small initial angles. For larger angles, the period clearly increases, reflecting the non-linear nature of the pendulum. The RK4 simulations for $\Delta t = 0.01$ and 0.001 precisely match the exact numerical integration curve. As predicted, the series expansion correctly models the increasing period up to a point, but noticeably diverges from the exact numerical solution as θ_0 approaches π , validating the non-linear theory.

Q2: Damped Pendulum

Method: We simulated the damped pendulum equation $\ddot{\theta} + \gamma\dot{\theta} + \omega_0^2 \sin(\theta) = 0$ using the RK4 method. For the first analysis, we plotted the natural logarithm of the positive peak amplitudes

against time to find the decay constant λ using linear regression. For the second analysis, we simulated the system across different damping coefficients γ and plotted the amplitude of the first turn to find when the system becomes overdamped.

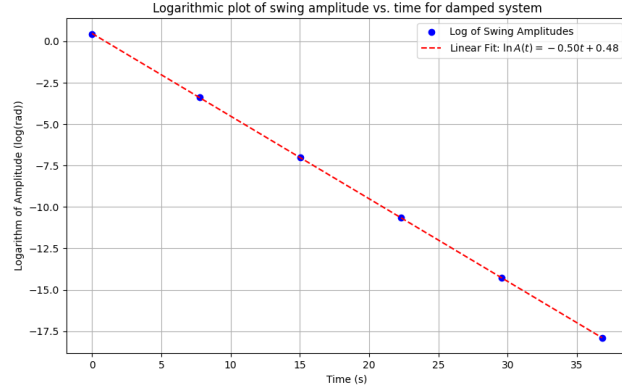


Figure 2: Logarithmic plot of swing amplitude vs. time for the damped system. Parameters: $\theta_0 = \pi/2$, $\gamma = 1.0$, $\omega_0^2 = 1$, $\Delta t = 0.01$, $t_{\max} = 40$.

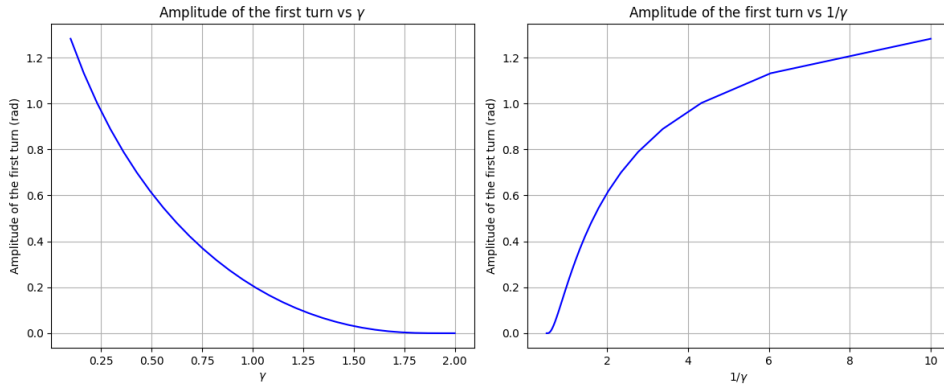


Figure 3: Comparison of the first turn amplitude for varying values of γ and $1/\gamma$. Parameters: $\theta_0 = \pi/2$, $\gamma \in [0.1, 2.0]$ (30 linearly spaced values), $\omega_0^2 = 1$, $\Delta t = 0.01$, $t_{\max} = 20$.

Theory: A weakly damped harmonic oscillator experiences an exponential decay in its amplitude over time, $A(t) = A_0 e^{-\lambda t}$. This implies that $\ln A(t)$ should decrease linearly with a slope of $-\lambda$. Furthermore, as the damping coefficient γ increases, the system transitions toward an overdamped state (which theoretically occurs at $\gamma \geq 2\omega_0$). In the overdamped regime, the pendulum should not oscillate or cross the equilibrium point, meaning its subsequent peak amplitude will be zero.

Results: Figure 2 shows a strong linear relationship for $\ln A(t)$ vs time, confirming that the nonlinear pendulum's envelope amplitude decays exponentially, just like a linear oscillator. The slope gives an estimated decay constant $\lambda \approx 0.50$. Figure 3 demonstrates that as γ increases, the amplitude of the first turn decreases. It reaches exactly zero at $\gamma = 2.0$, perfectly matching the theoretical threshold for critical damping ($\gamma = 2\omega_0$ where $\omega_0 = 1$), where oscillations are theoretically suppressed.

Q3: Driven Damped Pendulum

Method: We simulated the driven damped system $\ddot{\theta} + \gamma\dot{\theta} + \omega_0^2 \sin(\theta) = A \cos(\omega t)$ using the RK4 method. First, we measured the steady-state amplitude across various drive amplitudes A after allowing transients to decay. Second, we generated a phase portrait (θ vs p) for a specific high drive amplitude $A = 1.0$ by discarding the initial transient segment.

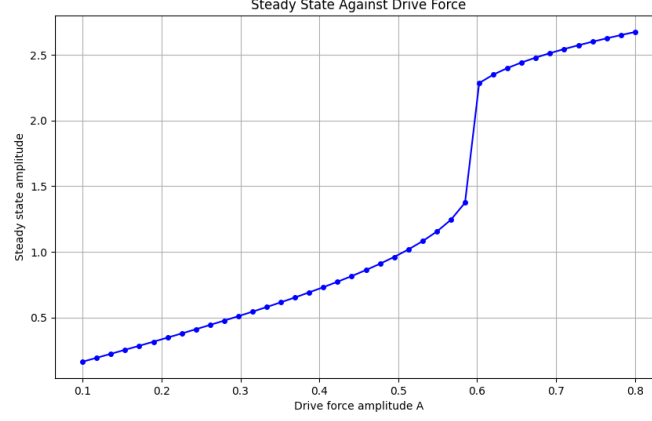


Figure 4: Steady state amplitude against drive force amplitude A . Parameters: $\gamma = 3/8$, $\omega = 2/3$, $\omega_0^2 = 1$, $A \in [0.1, 0.8]$ (40 linearly spaced values), $\Delta t = 0.01$, simulated until $t = 1000$.

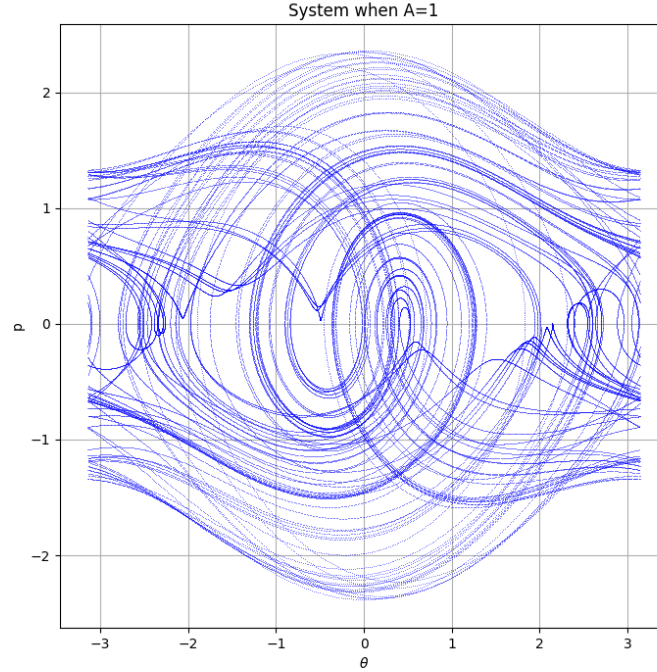


Figure 5: Phase portrait of the system when $A = 1$. Parameters: $\gamma = 3/8$, $\omega = 2/3$, $\omega_0^2 = 1$, $A = 1.0$, $\Delta t = 0.01$, simulated until $t = 1000$, transients discarded for $t < 200$.

Theory: For small driving forces, the pendulum is expected to settle into a stable periodic limit cycle with an amplitude proportional to A . However, for larger driving forces, the non-

linear term $\sin(\theta)$ becomes dominant. Theoretical dynamics implies this can lead to abrupt shifts in the steady-state response (due to resonance folding) and eventually chaotic behavior. In a chaotic regime, the phase portrait should not display a simple closed limit cycle, but rather a complex, fractal-like structure known as a strange attractor.

Results: In Figure 4, we observe the expected roughly linear increase in steady-state amplitude for small A , followed by a sudden jump around $A \approx 0.6$. This abrupt change is a clear indicator of nonlinear resonance effects. In Figure 5, the phase portrait for $A = 1.0$ is highly irregular and densely fills a region of phase space instead of forming a closed periodic loop. This confirms that the strong driving force has pushed the system out of predictable periodic motion and into deterministic chaos, exactly as nonlinear dynamics theory predicts.